

Curriculum Vitae: Elias Huseby Mathews.

Focuses: Scientific Machine Learning, Fluid Dynamics, Biophysics

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Professional Positions

March 2025 – Present
Picture Perfect Soft Matter Lab;
Methods: Machine Learning, CFD, HPC

Research Assistant, Stanford University
Advisor: Sho Takatori
Papers: 2nd Author [2]

September 2024 – March 2025
Environmental Complexity Lab;
Methods: Machine Vision, Particle Tracking

Research Assistant, Stanford University
Advisor: Nicholas Ouellette

February 2023 – September 2024
Fluid Dynamics and Soft Condensed Matter Lab;
Methods: Machine Vision, Particle Tracking

Research Assistant, Wesleyan University
Advisor: Greg Voth
Papers: 1st Author [1][3]

May 2022 – September 2022
In Vivo Neural Circuitry Lab;

Research Assistant, UC Santa Cruz
Advisor: Yi Zuo

Education and Training

September 2024 – Present: *Stanford University*

PhD Candidate, Environmental Fluid Mechanics; Mentor: Sho Takatori
MS, Computational and Mathematical Engineering: 3.69/4.0

February 2021 – May 2024: *Wesleyan University*

BA, Physics: 3.96/4.0

Awards and Honors

Phi-Beta-Kappa Honor Society (Wesleyan University)
Physics Department High Honors (Wesleyan University)
Bertman Prize for Excellent Physics Research (Wesleyan University)
Leroy-Apker Physics Prize Nominee (American Physical Society)

Teaching and Outreach

Teaching Assistant, *Wesleyan University*: Multivariable Calculus, General Physics, Advanced Classical Mechanics, Nonlinear Dynamics

Guest Speaker, *Gateway School*: Invited talks titled “Fluid, Solid, or Both!” and “How Scientists Use Measurements to Talk to Strangers”

Conferences

[Poster] Conference on Physics and AI (2026)
“*Advection Heads in an Atmosphere Foundation Model*”
[Poster] Berkeley Statistical Mechanics Conference (2026)

“Linear Hydrodynamics from Agent Based Models”

[Talk] American Physical Society Division of Fluid Mechanics (2023)

“Sedimentation Dynamics of Helical Ribbons”

Publications

Submitted/Under Review:

[3] **Huseby, Elias**, et al. “Bifurcations in Stokes Flow Sedimentation.” Submitted to Journal of Fluid Mechanics

Relevant Research Areas: Particle Transport, Nonlinear Dynamics

Published/In Press:

[2] MacMillan, Theodore. **Huseby, Elias**. Takatori, Sho. Ouellette, Nicholas “Advection Heads in an Atmosphere Foundation Model.” (PAI 2026)

Relevant Research Areas: Scientific Machine Learning

[1] **Huseby, Elias**, et al. "Helical ribbons: Simple chiral sedimentation." Physical Review Fluids 10.2 (2025): 024101.

Relevant Research Areas: Particle Transport, Nonlinear Dynamics

Mentorship

Wesleyan University, Physics: Mentored undergraduate students Meera Das (now post-bacc at Columbia), Arjun Menenez (Class of 2027), and Pierre Mathier (Class of 2027), in experimental, numerical, and theoretical fluid mechanics techniques.

Stanford University, Chemical Engineering: Mentored 1st year PhD student Sofia Rivalta Popescu (Chemical Engineering) in techniques for scientific machine learning, mechanistic interpretability, and computational/theoretical partial differential equations.